



AURA

WEB APPLICATION

THE CORE FOUR: SYDONYA, NCHIMUNYA, EMILY, BAL

MEET OUR TEAM



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Project Manager



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Full stack Developer



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Frontend Developer



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SYSTEM OVERVIEW

AURA (Adaptive User Reflective Assistant) is a comprehensive productivity application that integrates multiple learning and organization tools that is combined with proven psychological methods and studies.

AURA aims to provide students with a centralized platform that is curated to them in order to manage all aspects of their academic work while also incorporating proven learning techniques to enhance retention and understanding.

WHY THIS PROJECT?

We want to provide students with not just another study tool, but a personal learning assistant that grows with them throughout their academic journey, making evidence-based learning techniques accessible and engaging for everyone.



SYSTEM PLANNING & IMPLEMENTATION

REQUIREMENTS ELICITATION

We decided that observation would be the most efficient tool for the scope of our project. We observed productivity based web applications with similar features (like AI-powered systems) that we were able to get real-time insights from.

FUNCTIONAL REQUIREMENTS

Process Functional Requirements:

- The system must support content upload (PDF) of notes.
- The system must allow users to be able to create, edit, and organize flashcards by courses, subjects, and topics.
- The system must allow users to input their semester exam dates and generate a tailored study schedule with appropriately spaced quizzes and review sessions.

NON-FUNCTIONAL REQUIREMENTS

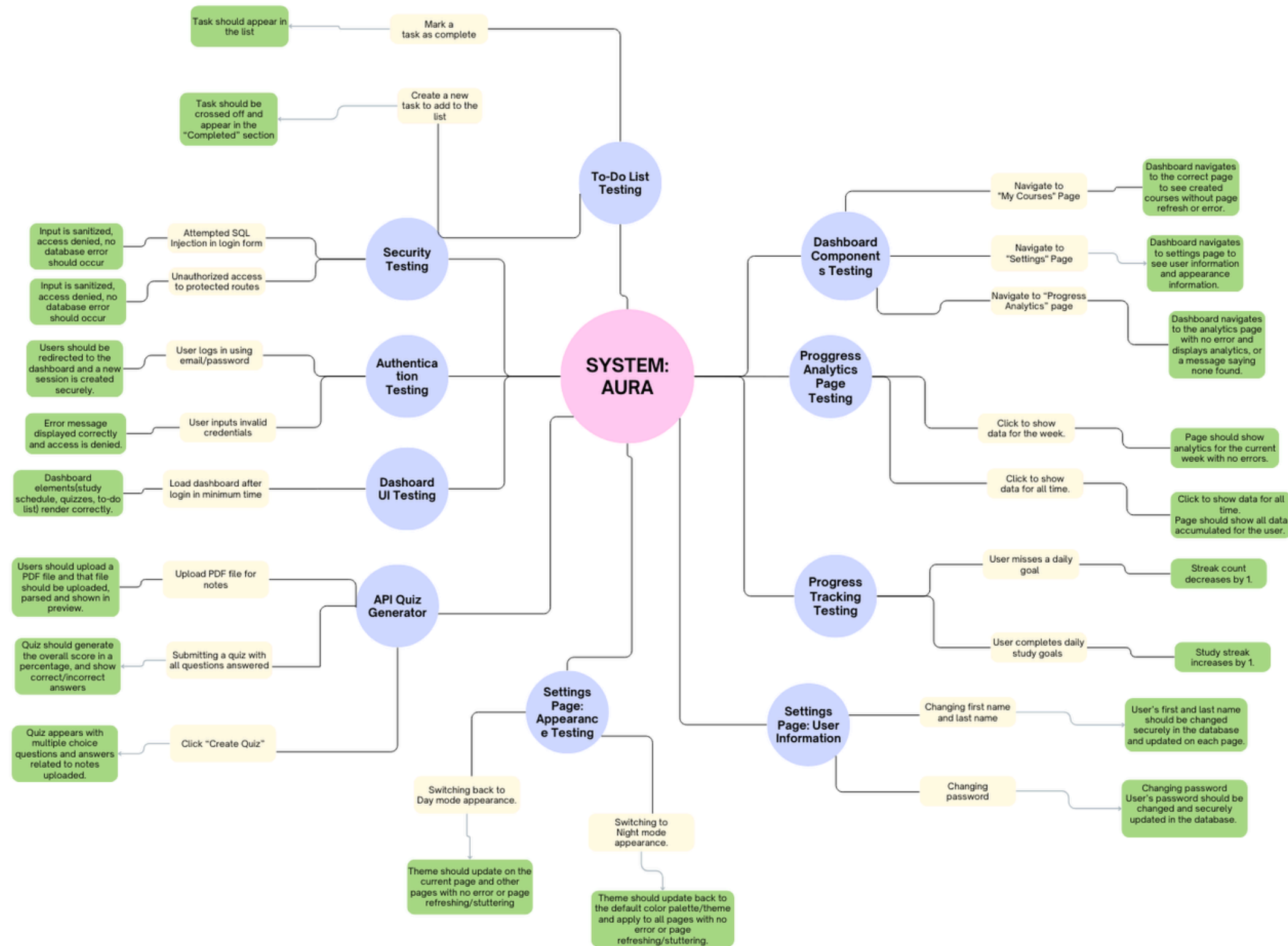
Performance Requirements:

- The system should be able to generate AI-powered quizzes and flashcards within 5 seconds of user request under normal load conditions.

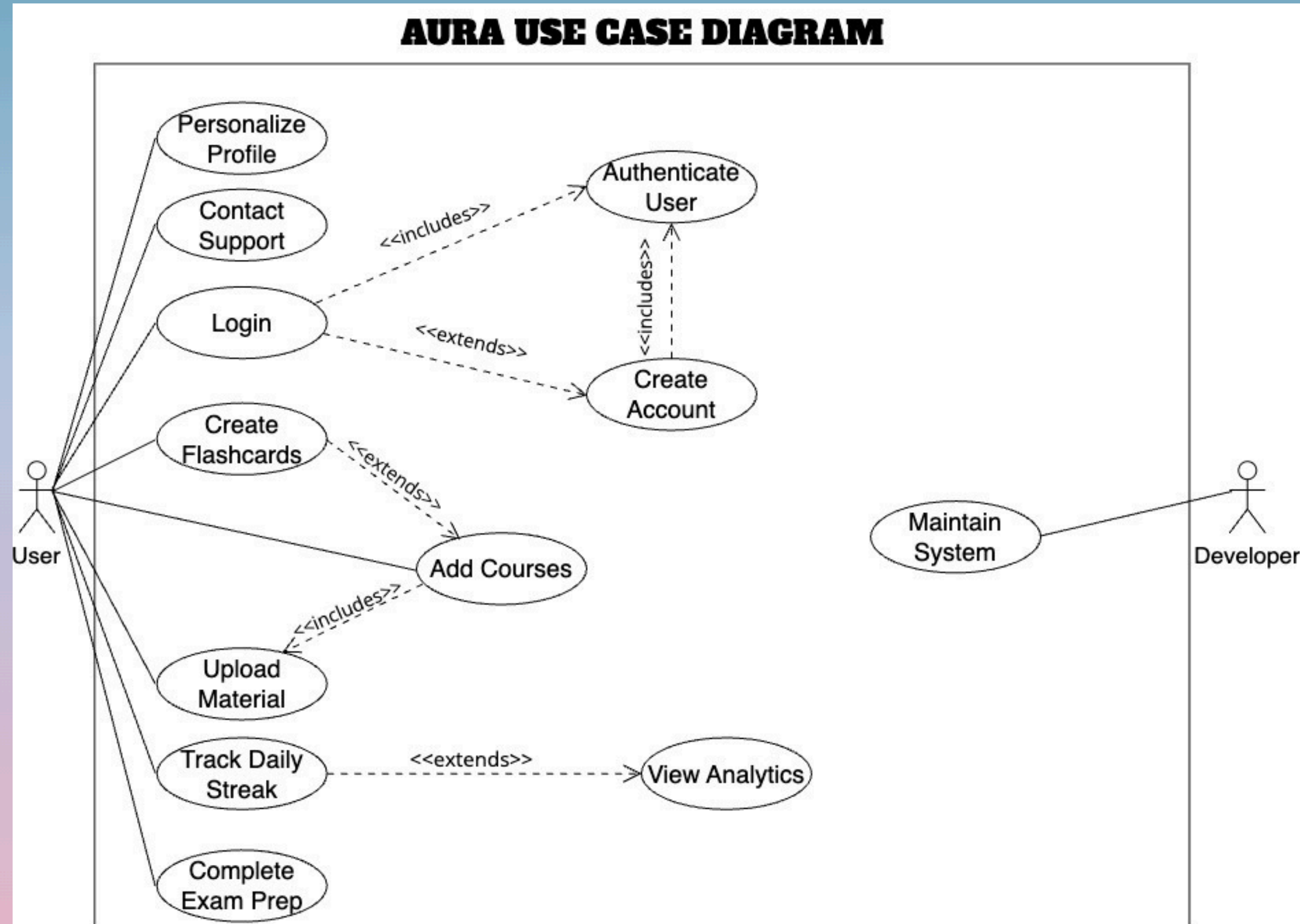
Dependability Requirements:

- The system should maintain 99.9% uptime during peak usage periods, such as exam preparation weeks and the start of each semester, to ensure consistent student access.

MASTER TEST PLAN

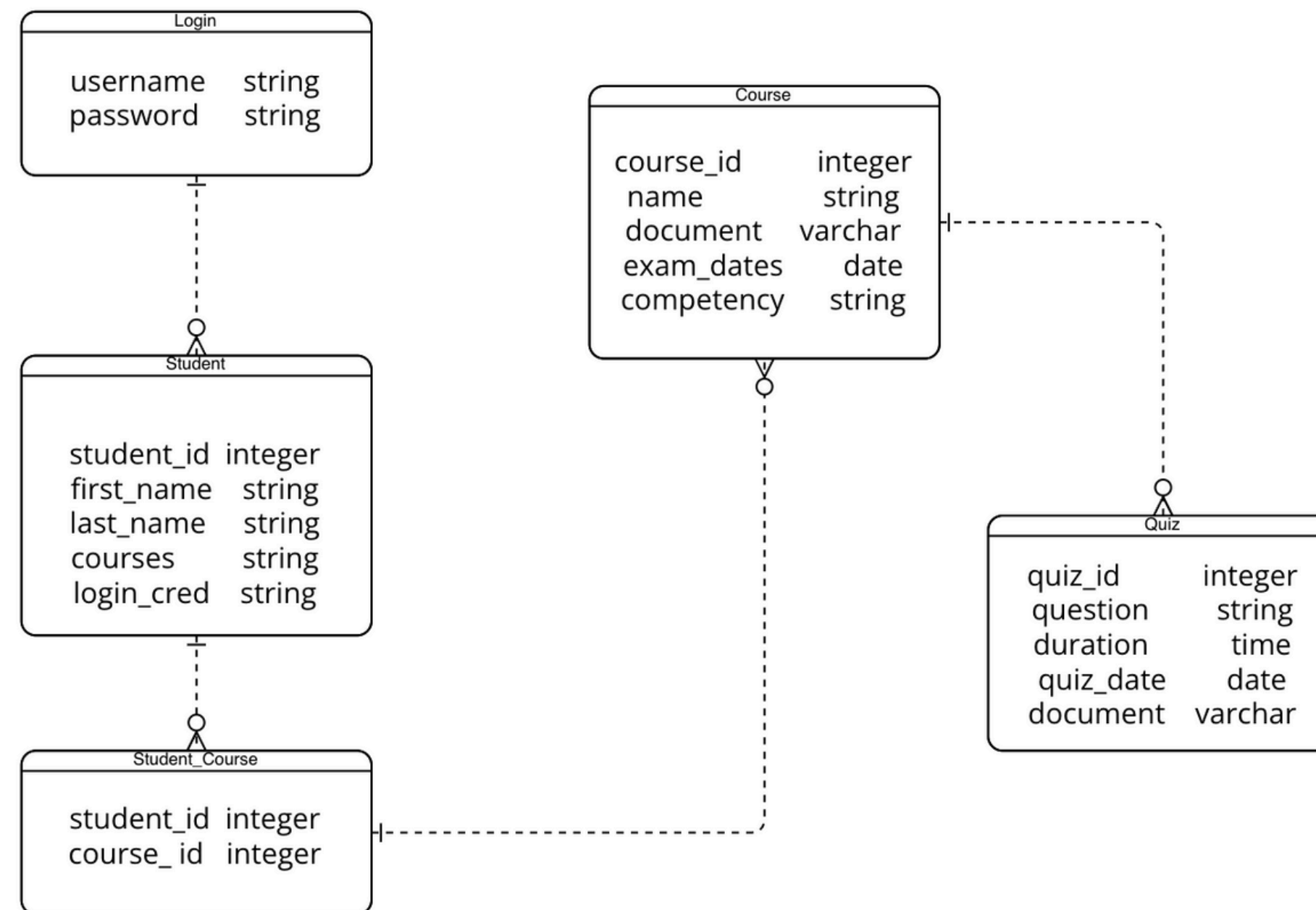


LOGICAL DESIGN



PHYSICAL DESIGN

AURA Entity-Relationship Diagram



SECURITY CONSIDERATIONS

User Authentication and Authorization:

- We established role-based access using Supabase.

SQL Injection:

- We implemented input validation before database queries were processed.

ChatGPT API Considerations:

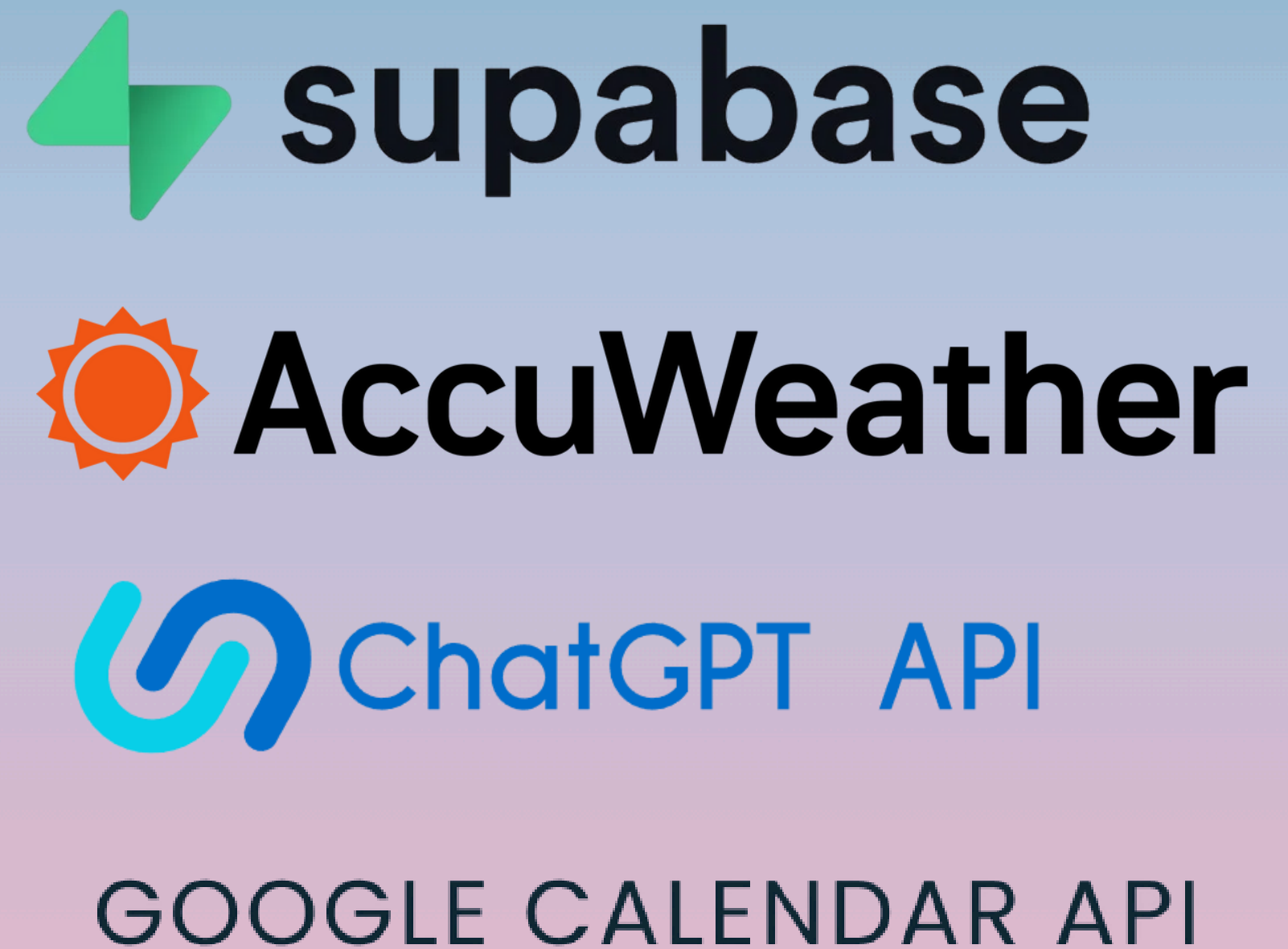
- We closely monitored the request rate to prevent too much charges.

TECHNOLOGIES USED

Frontend



Backend



CODE HIGHLIGHTS

ChatGPT API

```
const response = await fetch(
  "https://api.openai.com/v1/chat/completions",
  {
    method: "POST",
    headers: {
      "Content-Type": "application/json",
      "Authorization": `Bearer ${OPENAI_API_KEY}`
    },
    body: JSON.stringify({
      model: "gpt-3.5-turbo",
      messages: [
        {
```



DEMO

TESTING RESULTS

Authentication Testing

User logs in with username/password.

User inputs invalid credentials

Notes: All tests passed successfully.

API Quiz Generator Testing

Upload a
.txt file
for notes

Upload a PDF
file for notes

Click “Start
Practice
Quiz”

Submit a quiz with
only 1 question
answered.

Submitting a quiz
with all questions
answered.

Notes: All tests passed successfully.

HOW DO WE FIT IN THE MARKET?

Competitor Analysis				
FEATURE				
Target Users	✓	✓	✓	✓
Spaced Repetition Algorithm	✓	✗	✓	✗
Flashcard Generation	✓	✓	✓	✗
Progress Tracking	✓	✗	✗	✗
Quiz Generations	✓	✓	✗	✓
User Interface	✓	✓	✓	✓
Reusability	✓	✓	✓	✗



REFLECTIONS

OVERCOMING OBSTACLES

Learning new Technologies:

- We started by learning programming languages that were new to us before we began writing code.

Integrating the back-end:

- We used Supabase because it was a simple and easy database to implement.

Feature Creep:

- We added a lot of little features along the way but stayed focused on our major features.

WHAT ARE WE MOST PROUD OF?

- We are solving a real world problem.
- Each team member learned and mastered a new technology.
- **AURA** has a user-centered design.
- We were able to develop an intuitive spaced repetition algorithm that adapts to individual learning patterns.

WHAT DID WE LEARN?

- Documentation goes a long way.
- Preliminary planning is important.
- For a very complex project, prioritizing the core functionality should be high priority.
- Sometimes going with the initial instinct and idea for a project is a good idea.

CONCLUSION

- What we Built?
 - AURA is a personalized, AI-integrated study assistant that empowers and encourages users to have better study habits and increase productivity.
- Why it Matters?
 - AURA simplifies and strengthens the process of studying/productivity using automation, reusability, and science-backed methods.
- What's Next?
 - AURA will continue to be refined with its algorithm and be potentially expanded with mobile app/mirror functionality.



Q & A



THANK YOU!